One-Handed and Off-Balance: Does Proximity Feedback Modality Reduce Falls in a Smartphone Tilt Task?

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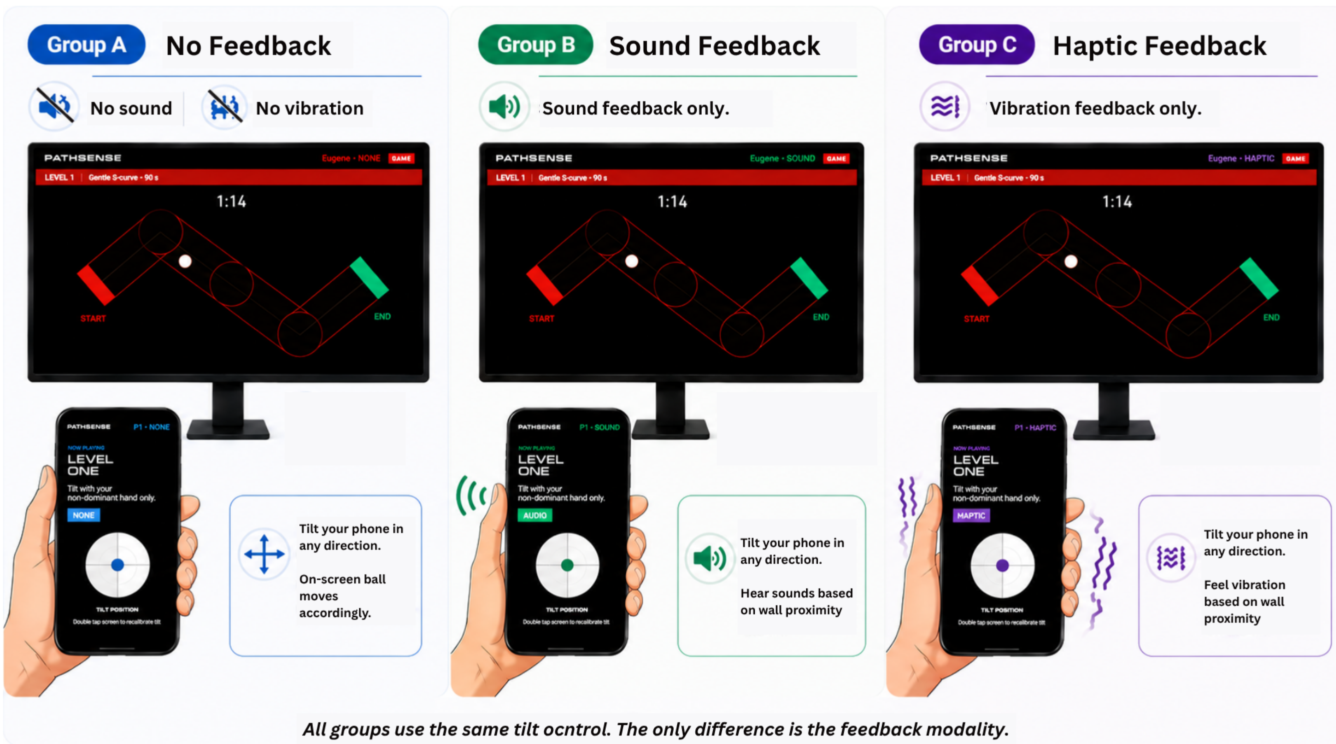

Figure 1: The three between-subjects feedback conditions in PathSense. All groups use the same tilt control. The only difference is the feedback modality.

Abstract

Tilt-based smartphone interfaces are ubiquitous, yet they often feel clumsy when controlled with the non-dominant hand. We hypothesize that this reflects a sensory calibration gap that proximity-based feedback could help close. We present PathSense, a smartphone-native tightrope tilt task comparing three feedback modalities (haptic, audio, none) with 25 participants across four difficulty levels. In this preliminary study, no modality showed a significant advantage in fall reduction (Kruskal-Wallis, all $p > 0.19$). However, change-from-baseline analysis reveals a critical confound in baseline practice performance (audio: 0.33 falls vs. haptic: 4.88, $p < 0.05$) that warrants controlled follow-up. We discuss implications for modality design in motor augmentation and provide effect size estimates for a larger planned follow-up ($n \approx 50$).

CCS Concepts

• **Human-centered computing → HCI theory, concepts and models**; *Haptic devices.*

Keywords

body schema, haptic feedback, motor learning, non-dominant hand, proximity feedback, smartphone, tilt interaction

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1 Introduction

Consider three everyday situations where one hand is occupied and the other must compensate alone: an elderly person pulling a shopping trolley while trying to navigate a smartphone with their free hand, a surgeon piloting a robotic arm through delicate tissue where a single mistilt causes harm, and a drone operator managing two control axes simultaneously under time pressure. In each case the same problem appears: one hand is busy, and the other hand is effectively blind. The off-hand receives no feedback about where it is or what it is doing, and without that feedback, the brain cannot adapt.

This difficulty has a neurological basis. Guiard (1987) described the non-dominant limb as the *frame-setter* in bimanual action, responsible for coarse spatial positioning rather than fine movement [1], and subsequent work has confirmed that the non-dominant hand relies more heavily on peripheral proprioceptive feedback than its dominant counterpart [2]. Crucially, the brain extends its spatial self-model into actively held tools [3], and recent evidence suggests smartphones specifically engage this body schema mechanism [4]. If the held phone is already encoded as a bodily extension, proximity warnings delivered *through* it may update the user's spatial model more directly than warnings arriving on a separate channel.

The open question is which feedback channel works best. Haptic cues act on the same sensorimotor channel as the task itself, while audio cues are processed separately and require deliberate signal-to-correction mapping [5]. Vibrotactile feedback has shown promise in proprioceptive rehabilitation [6], but whether smartphone-delivered proximity feedback produces comparable effects in a non-dominant hand tilt task has not been tested.

We present *PathSense*, a smartphone tightrope tilt task in which participants navigate a ball along a narrow pre-designed path using only their non-dominant hand, receiving graded proximity warnings in one of three modalities (haptic, audio, or none), shown in Figure 1. Falls (ball-off-path events) are the primary outcome. We address four research questions: (RQ1*) does modality affect fall rate? (RQ2) does any feedback reduce falls vs. control? (RQ3) which modality performs better? (RQ4) does improvement generalise across difficulty levels?

2 Related Work

Motor augmentation through external feedback has its roots in cybernetics and human factors research. Sigrist et al. (2013) found that haptic feedback excels at rapid online correction due to its colocation with the acting limb, while auditory feedback may better support learning in tasks requiring complex mapping [5]. In rehabilitation, haptic guidance has shown consistent gains for proprioceptive training [6], yet audio feedback is effective in tasks with complex movement mappings [5]. Regarding body schema, Lin et al. (2022) provided behavioural evidence that smartphone use updates body schema similarly to classic tool integration [4], suggesting that feedback delivered *through* a held phone may have privileged access to this system. Prior work on non-dominant hand control has focused on bimanual coordination [1], but not on proximity warnings in smartphone tilt. PathSense bridges this gap.

3 Study Design and Method

Participants. We recruited 25 participants (mean age 14.9, SD 1.8, 9 female, 19 right-handed) with no prior tilt-control experience required. Participants were assigned in counterbalanced order to haptic ($n = 8$), audio ($n = 9$), or none ($n = 8$) groups, and they did not differ significantly on age ($p = 0.37$) or gaming experience ($p = 0.41$).

Apparatus and Task. Participants held an Android smartphone with their non-dominant hand and tilted it to keep a ball centred on a narrow path displayed on an external monitor. Lateral tilt (gamma) drove horizontal acceleration, and forward-back tilt (beta) drove vertical acceleration. A *fall* occurred when the ball left the path corridor, after which the ball respawned at the most recent checkpoint.

Conditions and Procedure. The study was between-subjects (Table 1). Feedback escalated in real time as the ball approached the path edge: haptic pulses increased in intensity and frequency, audio tone pitch and rate escalated, and the none condition provided no feedback. After each round, participants completed a brief post-round survey.

Table 1: Difficulty levels and task parameters

Level	Path shape	Duration	Feedback
Practice (L1)	S-curve	90 s	None (all groups)
Easy (L2)	S-curve	90 s	Modality-dependent
Medium (L3)	N-shape	75 s	Modality-dependent
Hard (L4)	Tight zigzag	60 s	Modality-dependent

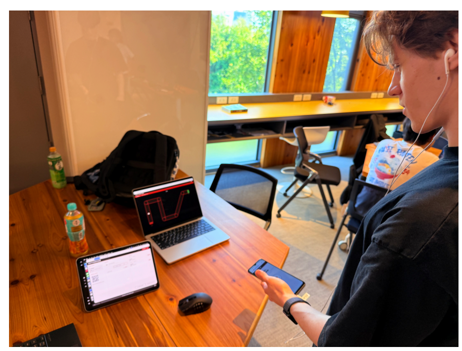

Figure 2: Study setup: participant holds the phone with the non-dominant hand and tilts to keep the ball on-path. The laptop displays the game in real time, and the tablet shows the admin panel.

Data Collection and Analysis. Game data were logged to an SQLite database via a Node.js server, and trajectory data were recorded at 10 Hz (ball position, proximity level, gamma/beta tilt angles). Rounds with missing `round_duration_ms` were excluded (4 of 112 expected rounds), yielding 100 rounds for analysis (25 sessions × 4 levels minus 8 excluded). Non-parametric Kruskal-Wallis H tests assessed modality effects at each experimental level, followed by pairwise Mann-Whitney U tests with Holm correction. Effect sizes are reported as rank-biserial r. Change-from-baseline deltas ($\Delta = \text{falls}_{\text{level}} - \text{falls}_{\text{practice}}$) controlled for heterogeneous baseline performance.

4 Results

Primary analysis: raw falls. Table 2 shows mean falls by modality and level. Kruskal-Wallis tests on experimental levels showed no significant modality effect: Easy $H(2) = 3.294$, $p = 0.193$. Medium: $H(2) = 0.600$, $p = 0.741$. Hard: $H(2) = 0.477$, $p = 0.788$. Pairwise Mann-Whitney tests (Holm-corrected) confirmed no significant differences at any level (all $p > 0.27$, rank-biserial r from -0.44 to $+0.25$). To quantify evidence for the null, Bayes factors strongly favour no modality difference: $BF_{01} \approx 83.8$ at Easy, 7.0 at Medium, 5.4 at Hard. A winsorised sensitivity analysis (95th percentile per cell) reproduced the same pattern (all $p > 0.16$).

Table 2: Mean (SD) falls by modality and difficulty level. Practice is the pre-feedback baseline (no feedback given to any group). Group sizes: Haptic $n = 8$, Audio $n = 9$, None $n = 8$.

Level	Haptic ($n = 8$)	Audio ($n = 9$)	None ($n = 8$)
Practice	4.88 (7.34)	0.33 (0.50)	3.00 (4.17)
Easy	3.50 (5.71)	0.33 (0.71)	0.62 (0.74)
Medium	3.62 (6.39)	2.44 (3.81)	3.62 (6.32)
Hard	13.00 (4.04)	11.78 (7.36)	13.25 (5.18)

Baseline confound. A critical asymmetry emerged in the practice round (L1), where no feedback was given: audio participants averaged 0.33 falls (SD 0.50) while haptic averaged 4.88 (SD 7.34) and none averaged 3.00 (SD 4.17), representing a 10-fold gap (Mann-Whitney $U = 28$, $p = 0.04$, $r = -0.57$). This suggests pre-existing group differences in baseline tilt control. Change-from-baseline deltas show that at Easy level haptic improved by -1.38 falls, none by -2.38, and audio remained flat (0.00). At Hard level, all groups deteriorated by $+8$ to $+11$ falls with no modality advantage.

Steering response. To investigate real-time responses to proximity warnings, we identified 2,372 proximity escalation events (consecutive frames where `proximity_level` increased) and extracted a ± 4 s window around each, excluding events within 500 ms of a fall or respawn. Figure 3 shows time-locked mean proximity level and tilt magnitude per modality. Median time-to-recovery was similar across groups: haptic 400 ms, audio 416 ms, none 417 ms, and 98–99% of escalations recovered within 3 s. Tilt reversal frequency within 500 ms differed slightly (audio 43.0%, none 32.6%, haptic 27.2%), yet all modalities showed comparable proximity return to baseline within 1–2 s.

Qualitative feedback. Participant P008 (audio) noted: *"The sounds made me nervous. It's a good training for elders or students with special needs."* This is consistent with audio cues increasing arousal without improving control.

5 Discussion

A null result that points elsewhere. PathSense found no significant fall-rate differences across modalities at any difficulty level.

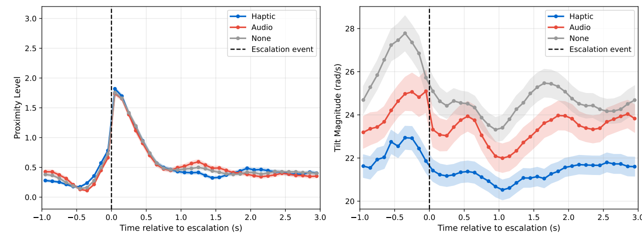

Figure 3: Time-locked proximity level (left) and tilt magnitude (right) relative to proximity escalation events. Vertical dashed line marks the escalation moment ($t = 0$). Events within 500 ms of a fall or respawn were excluded. Haptic (blue), Audio (medium gray), None (dark gray). Shaded bands: ±SE.

Bayesian analysis confirms this is genuine evidence for the null, not merely insufficient power: $BF_{01} \approx 84$ at Easy level is far above the conventional threshold for strong null evidence, and effect sizes were negligible (rank-biserial r from -0.44 to $+0.25$). Saying "no modality wins" is itself a useful finding for HCI, because it tells us that choosing the right feedback channel is not what determines performance [7].

Baseline and difficulty matter more than modality. The more informative result emerged before any feedback was given. At the practice round, the audio group averaged just 0.33 falls while the haptic group averaged 4.88, a 10-fold gap ($p = 0.04$, $r = -0.57$) driven entirely by pre-existing differences in tilt control ability. This is the real story: individual baseline responsiveness varied dramatically, and that variation outweighed anything the feedback modality contributed. Change-from-baseline analysis reveals moderate modality-specific patterns (Cohen's $d \approx 0.5$–0.7), with haptic and none groups improving at Easy while audio remained flat. Schmidt and Wulf (1997) showed that continuous concurrent feedback can degrade rather than support skill acquisition [8], which may explain why real-time audio escalation cues did not translate into fewer falls. At Hard level, task difficulty overrode everything, with all groups deteriorating by $+8$ to $+11$ falls regardless of modality. Taken together, these findings suggest that the right question is not which feedback technology to deploy, but rather how to measure a person's baseline control ability first and then design feedback to match it.

6 Limitations and Future Work

With 8–9 participants per cell, the study is underpowered. A follow-up with $n \approx 50$ per modality would achieve 80% power for medium effects. Most participants were 13–15 years old, while the theoretical background draws on adult motor control and body schema literature. Replication with adult samples is essential to separate age effects from modality effects. Random assignment did not equate practice-round performance, so future studies should counterbalance or control the practice phase to avoid the baseline confound observed here. Finally, each participant completed only one session, so learning consolidation and retention were not assessed. Multi-session designs are needed to determine whether modality differences emerge with extended practice.

7 Conclusion

PathSense demonstrates a smartphone-native platform for comparing proximity feedback modalities in a tilt-controlled motor task. In this pilot study we found no raw differences in fall rate across modalities, but uncovered a critical baseline confound and evidence that change-from-baseline analysis reveals modality-specific patterns (haptic and none improve at easier levels, while audio remains flat). These findings suggest that feedback modality effects may be subtle, task-dependent, and developmentally sensitive. Rather than declaring haptic a clear winner, this work highlights the importance of (i) controlled baselines, (ii) developmental-stage matching in theoretical predictions, and (iii) larger sample sizes for confirmatory claims. The PathSense platform and pilot data provide effect size estimates and design specifications for a larger follow-up.

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